

Leveraging GNN-Driven Inference with Pub/Sub Messaging for Coherent Data Erasure in Multi-Node Cloud Systems

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Abstract

Large-scale, distributed data storage is made possible by cloud computing, which also improves system performance and availability. However, a recurring and unsolved problem is making sure that data is consistently and completely deleted across several servers. Copies of a file that has been deleted by a server might still be present on other network nodes, which could result in data leakage, violations of privacy laws, and wasteful storage resource usage. In order to resolve the discrepancy, this work presents an intelligent system that combines Graph Neural Networks (GNNs) with a Publish/Subscribe (Pub/Sub) messaging to make this deletion. The entire storage infrastructure is modeled as a dynamic graph, where servers and data blocks are represented as nodes, and replication relationships as edges, enhanced with attributes like last access time, trust levels, and file size. When data deletion takes place, the Pub/Sub model notifies all subscribed nodes, which then use GNN to predict other possible data locations. Based on the prediction confidence, the system requests admin approval, or deletes the data automatically. This approach is highly scalable, and reduces duplicate actions and helps in proper detection of the deleted data from other servers. Integrated auditing feature which is similar to blockchain, provides a secure and record of all deletion operations, increasing liability. The proposed framework enables reliable, efficient, which is really crucial in data-sensitive sectors such as healthcare, finance, and education. Its design supports both operational efficiency and adherence to privacy regulations across distributed storage environments. Thus this approach has been taken.

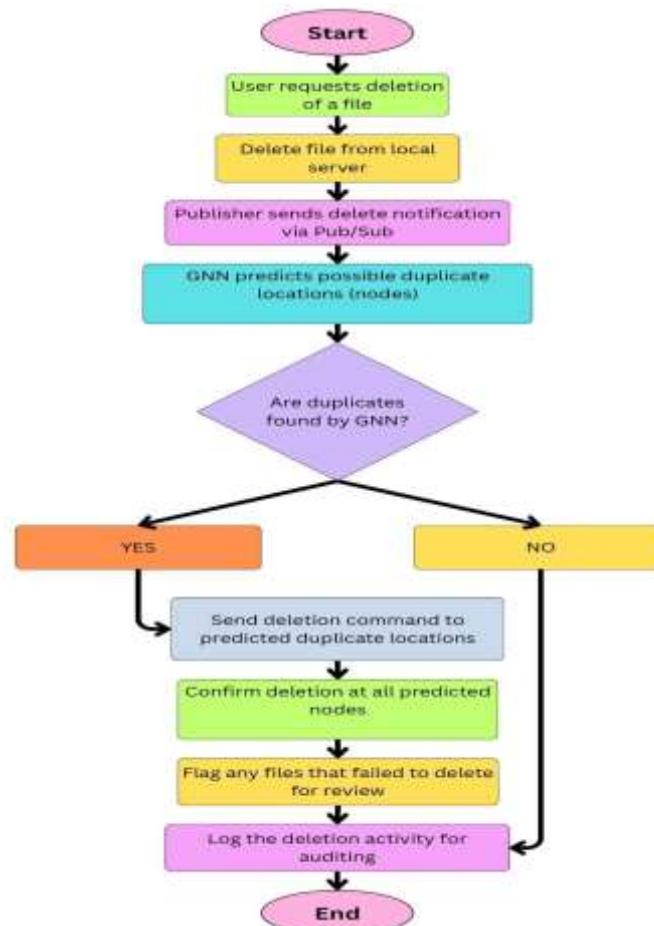
Keywords: Cloud computing, Publisher, Subscriber, GNN, Blockchain, Data deletion, PUB/SUB, Graph Neural Network.

1. Introduction

As cloud computing has become one of the most important methods for modern data storage and processing, the need to ensure data privacy, protection, and compliance with legal standards has become more important than ever. Some major cloud service providers, such as Amazon Web Services (AWS) and Google Cloud Platform (GCP), store multiple copies of the same data in different data centers across the globe. To improve system performance, increase reliability, and add redundancy in case of failure it is

done[5, 6]. In some ways duplication helps but, it also introduces certain risks especially for private, sensitive, or legally protected data. When a file is deleted from one server, it doesn't necessarily mean that every copy of it is removed from all locations. This can lead to privacy issues or violations of data protection laws [2, 4]. To solve this issue, we introduce a system that brings together the Publish/Subscribe (Pub/Sub) messaging model and Graph Neural Networks (GNNs). This combination aims to guarantee that all duplicate data is safely and fully deleted from the system [7]. In our design, the Pub/Sub model plays a central role. Servers are grouped into "publishers" and "subscribers." When a change, such as data deletion, happens, the publisher server sends out a message to alert all the other servers [3]. These other servers—now acting as subscribers—receive the alert immediately. This real-time notification helps ensure that every server holding a version of the file becomes aware of the deletion request. As a result, the chances of leftover or orphaned data copies being forgotten in the system are greatly reduced [1]. In addition to that, Graph Neural Networks are used to help locate all possible duplicates of the data across the network. In our system, each data file and each server is considered a node in a digital graph [6, 8]. The relationships—or edges—between these nodes show how data flows, how often it is accessed, how similar files are to each other, and how trusted the storage locations are. Even small changes to the system or file access patterns are picked up by the GNN, which is designed to learn from this behavior. Over time, the model becomes better at predicting where extra data copies might still exist [4]. The system finds which servers need to be updated or cleaned and makes the deletion process faster and smarter. The process can either handle this process automatically or with help from a human administrator. The Pub/Sub model ensures quick and

Figure1: Working Flowchart Diagram of the model



dependable communication between servers, while the GNN helps the system adapt and improve its predictions as new data gets updated[8]. As it is properly shown in Figure. 1 the flowchart which shows how the working of the whole system and this combined approach of GNN and pub/sub improves both the security of stored data and the efficiency of storage usage. More importantly, it helps cloud systems follow global data protection laws and meet the needs of industries that depend on strong data privacy—such as healthcare, banking, and education.

2. Working Principle

In this system, there are two main technologies used: the Publish/Subscribe (Pub/Sub) model and Graph Neural Networks (GNNs)[3, 7]. This guarantees that when a file is deleted then all its duplicate copies are securely and completely removed and gets no duplicated from the cloud [5].

First, let us talk about the Pub/Sub model. In a cloud system data is copied in-between different servers in different locations [1][6]. When a user deletes a file, it is not enough to remove it from just one server because the same file might still exist in other places [4]. The Pub/Sub model helps in sending a message to all the other servers whenever a file is deleted or changed [3, 5]. In this model, servers are assigned with two roles: publishers and subscribers [2]. The server where the deletion happens acts as the publisher and sends out a message [8]. All the other servers that need to be updated about file changes act as subscribers and they immediately receive this message [1]. This way, the system ensures that all the servers are informed in real-time when something changes [4, 6]. Thus, no duplicate is left behind unknowingly. As an example, like a group message in a chat, when one person sends a message (the publisher) then everyone in the group (the subscribers) gets notified instantly and the subscriber sees it [5, 7]. The Graph Neural Network (GNN) which acts as an intelligent system that works with graphs where the structures are made up of nodes and edges [7, 9, 10]. Each server and data file is treated as a node and the connections between them (such as file sharing, movement, or access patterns) are the edges [9]. Figure 2 shows how the architecture is actually working [11]. The GNN studies how data moves around in the system and learns

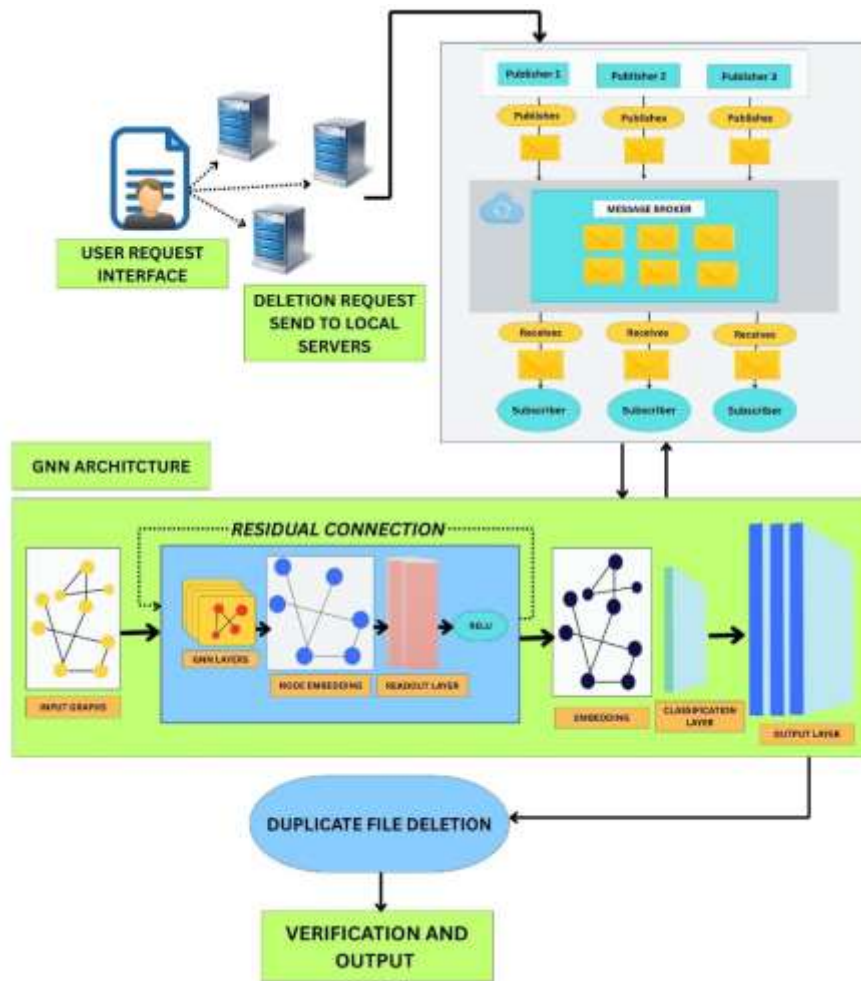


Figure2: Representation of whole architecture of the model

patterns [12, 18]. For example, the servers share files or store similar copies [10, 19].

The equation of Convolutional Neural Network:

$$h_v^{(l+1)} = \sigma \left(\sum_{u \in N(v)} \frac{1}{c_{vu}} \cdot W^{(l)} h_u^{(l)} + W^{(l)} h_v^{(l)} \right)$$

(1)

Where, $h_v^{(l)}$ is the hidden state (embedding) of node v at layer l , $N(v)$ is the set of neighbours of node v (servers/files connected to v), c_{vu} is a normalization constant (e.g., degree of node v , or a fixed value like $\sqrt{|N(v)||N(u)|}$), $W^{(l)}$ is the learnable weight matrix for layer l , which transforms the features, σ is a non-linear activation function, commonly ReLU, $h_v^{(0)} = x_v$, the initial node features.

So, when a file is deleted, the GNN can predict where other copies of that file might be located, even if those servers did not directly receive the file from the original source [18, 23]. It acts like a smart detective, looking at clues from the system to find hidden or indirect duplicates [14, 21]. When Pub/Sub combines with GNN, a system is formed that is both fast and smart [1, 7, 22]. The Pub/Sub model makes sure that any update is quickly sent to all servers, so everyone is aware [2, 5]. Meanwhile, the GNN looks deeply into the system and finds out where duplicates are likely to be, even if they were not directly notified [18, 19, 24]. Based on this, the system can either automatically delete the duplicates or flag them for review [9, 20]. This combination ensures that data is deleted from a single location as

well as from all possible locations, which makes the cloud system more secure, more private, and more efficient [4, 13, 25]. This also helps companies such as GDPR to follow strict data protection rules, because they can be sure that deleted data is really gone from everywhere [6, 22].

3. Advantages and Disadvantages

There are several advantages and disadvantages related to this work and some of them really points out the purpose of the work those advantages and disadvantages are mentioned below:

3.1. Advantages

3.1.1 Better Data Deletion Across All Servers: The deletion of a file on one server doesn't remove its copies elsewhere in cloud systems [1, 4]. The system uses a messaging method (Pub/Sub) to alert all connected servers about the deletion [2, 5]. This helps to make sure no copies are accidentally left behind [6].

3.1.2 Smart Detection Using Graph Neural Networks (GNNs): GNNs are like smart algorithms that understand connections [7, 9]. They can learn how files are shared or duplicated across servers [10]. If a file's location was not clearly recorded, then the GNN can guess where it might be and help to find and delete it [11, 18].

3.1.3 Works Well for Large Cloud Networks: Big companies like AWS or Google Cloud have thousands of servers. The system can easily scale to manage data deletion in such environments [5, 8]. The Pub/Sub model handles large numbers of servers efficiently and keeps communication smooth [2, 17].

3.1.4 Less Work for System Administrators: The system does not always need a human to tell it what to do [12]. The prediction mechanism allows it to choose to delete the file, ask for permission, or just make a note [13, 19]. This saves a lot of time and effort for the operator [20].

3.1.5 Helps Follow Privacy and Legal Rules: Privacy laws like GDPR and HIPAA require that user data can be fully deleted when requested [21]. The system tracks and guarantees deletion across all data copies [15, 5, 22]. So it supports compliance with these strict laws, which is very important in sensitive areas like finance and healthcare [24].

3.1.6 Reduces Storage Waste: When copies of the same file are left all over the system, there is wastage of valuable storage space [4, 6]. The system removes duplicates [3, 5]. It helps to free up space and reduce costs for cloud storage [8].

3.1.7 Permanent and Secure Deletion Records: This system keeps a protected log (like a blockchain) of every deletion event [1, 4]. This allows anyone to check what was deleted, when it was deleted, and by whom [5, 7]. Which is useful for audits, security checks, or legal purposes [6].

3.2. Disadvantages

3.2.1 Communication Needs Strong Security: The Pub/Sub system sends deletion signals, which without protective measures could be intercepted by a hacker to send fake commands for deletion [8, 23]. Hence, very high security and access control are prerequisites [5, 17].

3.2.2 Hard to Build and Set Up: The combination of Pub/Sub messaging and GNNs makes for a much more complex system [9, 12]. It needs thorough knowledge of cloud architecture, data science, and security [18]. So, essentially, this kind of system is hard to set up, with a lot of planning, coding, and testing involved in it [10].

3.2.3 Needs Computing Power: GNNs handle super large graphs like maps illustrating all servers and files [7, 19]. The hardware demand requirement for deletion in real-time is, therefore, very high [20, 23]. This increases system costs [24].

3.2.4 Depends on Information Being Good and Updated: The GNN model works well when good information is available detailing which servers hold which files, when last used, and how trusted each server is [9, 18]. The system will fail if there is insufficient data [19].

3.2.5 Possible Delays in Deletion Actions: The Pub/Sub system sends alerts quickly [2, 5], but the GNN still needs time to analyze the graph and make predictions [9][20]. This could delay the actual deletion of some files, especially in very large networks [23].

Conclusion

This system brings together the publish-subscribe (pub/sub) model and Graph Neural Networks (GNNs) to ensure that data is fully and securely deleted across the cloud. When a file is deleted, the pub/sub model immediately notifies all connected servers so they can respond at once [17]. This guarantees that no server is left unaware of the deletion. It helps in maintaining consistency across the system. At the same time, GNNs act like intelligent detectives—learning how data moves, how it's interconnected, where it's stored, and predicting where hidden or duplicate copies might exist [7, 9, 10]. As it detects the duplicate data, the system can delete them automatically or ask for manual confirmation, depending on the admin. Every deletion is logged clearly and immutably, similar to how blockchain systems work—creating a transparent, traceable record of all actions taken [1, 2, 5, 6]. This strengthens data security as well as ensures full compliance with data protection regulations like the GDPR [17]. This system combines real-time communication with intelligent data tracking and transparent logging. It also helps to prevent data from leaking, and reduces the risk of unauthorized access significantly reducing unnecessary storage usage [3, 4]. This smart approach goes beyond the traditional deletion methods by anticipating risks and acting swiftly [20, 23]. This method also holds promising potential for industries like healthcare, finance, and education—sectors where data privacy and compliance are non-negotiable [18, 22]. Thus, this kind of solution could transform how sensitive data is managed, monitored, and ultimately secured.

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